



High accurate lightweight deep learning method for gesture recognition based on surface electromyography

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ABSTRACT

Background and objectives: Surface Electromyography (sEMG) is used mostly for neuromuscular diagnosis, assistive technology, physical rehabilitation, and human-computer interactions. Achieving a precise and lightweight method along with low latency for gesture recognition is still a real-life challenge, especially for rehabilitation and assistive robots. This work aims to introduce a highly accurate and lightweight deep learning method for gesture recognition.

Methods: High-density sEMG, unlike sparse sEMG, does not require accurate electrode placement and provides more physiological information. Then we apply high-density sEMG, which, according to previous studies, leads to sEMG images. In this study, we introduce the Sensor-Wise method, which has a higher capability to extract features compared to the sEMG image method due to its high compatibility with the nature of sEMG signals and the structure of convolutional networks.

Results: The proposed method, because of its optimal structure with only two hidden layers and its high compatibility, has shown no sign of overfitting and was able to reach an accuracy of almost 100% (99.99%) when it was evaluated by CapgMyo DB-a database through 96 electrodes. Using this method, even with 16 electrodes, we were able to reach an accuracy of 99.8%, which was higher than the accuracies reported in the previous studies. Additionally, the method was evaluated by the CSL-HDEMG database, where the accuracy reached 99.55%. Previous studies either introduced expensive computational methods with overfitting or reported lower accuracies compared to this study.

Conclusions: The Sensor-Wise method has high compatibility with the nature of sEMG signals and the structure of convolutional networks. The high accuracy and lightweight structure of this method with only two hidden layers make it a proper option for hardware implementation.

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1. Introduction

Applications of Human-Computer Interfaces (HCI) in various industries are dramatically increased by progressing technology. Among the different developed techniques for HCI, those based on electromyography (EMG) are widely used for hand gesture recognition to control rehabilitation equipment [1, 2], industrial robots [3, 4], and also in new industries such as virtual reality as well as augmented reality [5, 6]. EMG is an electrodiagnostic medicine technique, which is used to record and evaluate the electrical activities of muscles. These electrical changes are the result of signals emitted from the brain to the muscles [7]. At the end of the brain-

muscle pathway, there are motor units, which transmit signals to the contractile protein to produce muscle force. The discharge or firing of each motor unit corresponds to an action potential, which is applied to all fibers composing the motor unit. A muscle consists of many motor units. The characterization of the generated activation signal for the intended muscle can be extracted from the sum of the action potentials corresponding to these motor units [8].

Given that most of the finger muscles are located in the forearm area, it is possible to record movements of the fingers by placing electrodes in the forearm area instead of the fingers [9]. There are two types of EMG sensors: intramuscular EMG (imEMG) and surface EMG (sEMG). The imEMGs are actually needles, which are directly inserted into the muscle to record electrical activity. Although the signal-to-noise ratio is better in the imEMG method, this method has its obstacles due to its aggressive nature [10].

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In contrast, surface electromyography (sEMG) is a non-invasive method, which is used to record muscle's electrical activity from the surface on (not into) the skin. The amplitude of sEMG not only depends on the potential received from the nervous system but also relies on the properties of fibers and skin [10]. Due to the non-invasive nature of sEMG, most recent studies of hand gesture recognition are based on sEMG electrodes [11, 12].

There are two approaches for applying sEMG electrodes over a limb: (1) sparse multichannel sEMG and (2) High-Density sEMG (HD-sEMG). Sparse multichannel sEMG usually requires precise positioning of electrodes [13–15], whereas, in HD-sEMG, two-dimensional arrays of electrodes cover a significant area of a limb that precise electrode placement is not required. Due to the limitation of sparse multichannel sEMG and capability of HD-sEMG over providing better spatiotemporal characteristics, muscle computer interfaces are recently based on HD-sEMG [16–18].

Gesture recognition method entirely depends on the nature of the input data. According to the non-uniform distribution of innervation numbers across motor units, which form a muscle, the number of fiber action potentials that involve an EMG signal is not directly related to the amount of motor unit action potentials. In addition, according to the placement of the sEMG electrode, more than one muscle may contribute to the consequent signal. Both these criteria cause non-linear and non-stationary nature of sEMG signals [8, 19].

Artificial neural networks (ANN) provide reliable methods for modeling the complex non-linear relationships between inputs and outputs based on statistic data, which are provided to them [20]. Gesture recognition based on ANNs is addressed as a classification problem where classifiers are trained through supervised learning. Machine learning classification methods such as Multi-Layer Perceptron (MLP), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forests, and Linear Discriminant Analysis (LDA) are used widely and even conventionally for non-invasive gesture detection. These methods often reported average accuracies up to 80–90% [21–23]. Although deep learning methods presented better accuracy in comparison with mentioned machine learning methods in recent studies but these methods, also face challenges.

Many gesture recognition applications, especially controlling rehabilitation or assistive robots require a real-time system response. Real-time systems with low latency ratio not only increase the performance of the robots at high speeds and make the movements look more natural, but also they have a significant effect on systems safety. On the other hand, since most of this equipment is mobile, embedded systems are used to build them, rather than general-purpose systems. Embedded systems are specifically designed and built for the intended use; therefore, they should be optimized in terms of efficiency, power consumption, and dimensions. Computational cost and subsequently, hardware resources cost are considered as the main concerns for developing real-time embedded systems. The computationally expensive methods require more hardware resources, and the latency rate of these methods is high [24, 25].

Although deep learning methods have shown promising results in recent studies, they still face challenges [21, 26]. The previous reports either applied many layers hence became computationally expensive or have faced overfitting problems. The presence of a layer and even in some cases multiple layers of dropout in these networks indicates a significant overfitting problem. Therefore, we introduced the Sensor-Wise method, which has a high capability to extract features due to its high compatibility with the nature of sEMG signals and the structure of convolutional networks. The method is lightweight and has a lower computational load because it has an optimal size with only two hidden layers, and there are not any batch normalization functions in the network structure. Because of its lightweight structure and high compatibility, it has

shown no sign of overfitting and was able to reach a higher accuracy even with fewer sensors compared to previous reports [21–23, 26–28].

2. Material and methods

2.1. Background

Convolutional neural network (CNN) as a deep learning method in many fields such as object detection, speech recognition, drug discovery, and hand gesture recognition has performed much better than other commercial methods [28, 29]. Many recent studies have demonstrated the benefits of using CNN in hand gesture recognition. In these studies, HD-sEMG is usually associated with sEMG images that their pixels include instantaneous values of all electrodes. Each image describes the spatial distribution of electrical potential by itself in the area where the electrodes arrays are applied. The sequence of these images can be interpreted to provide temporal information about changes over time [21, 26]. The data provided by the HD-sEMG images may be useful in describing changes in electrical properties in the pick-up area. However, the capability of this method in gesture recognition needs to be investigated more along with other methods.

Geng et al. achieved the accuracy of 99.5% using 128 electrodes and applying sEMG images to 8-layers network, which consists of two convolutional, two locally connected, and four fully connected layers along with many batch normalizations and three dropout layers [21]. Hu et al., achieved 99.7% accuracy at the expense of heavier network by upgrading the convolutional network that presented in the previous report (Geng et al.) and adding an LSTM layer [26]. Besides being computationally expensive, they did not achieve the accuracy of 100%, and the presence of multiple dropout layers with a weight of 0.5 indicates that the network has a significant overfitting problem.

Usually, a network is trained with training-data and evaluated with validation-data during training and eventually tested with test-data, which was not introduced to the network during the training process. In a network with an overfitting problem, the training error decreases steadily, but the validation error starts to increase from somewhere. Overfitting occurs when the network has more parameters than the optimal one or using a more complicated approach compared to the optimal approach. The nature of overfitting is that it extracts some unwanted changes (such as noise) as if these changes represent the basic structure of the model [30]. Using the dropout method is one way to reduce the overfitting effect by adjusting the output weight of some neurons to zero; in other words, it removes the trained cells from the next layers randomly.

Compared to Geng et al. and Hu et al., most of the recent studies have reported lower accuracy, and those who achieved similar accuracy are computationally expensive and suffer from overfitting [21–23, 26–28]. Peng Xia et al. introduced a network that consists of three CNN and two LSTM layers along with the dropout technique with a rate of 30%. Their network reached an accuracy of 94.7% [31]. Wei et al. used a multi-view deep learning network (denoted as MV-CNN) that includes multiple single-view networks (SV-CNN). Each single-view network consists of two CNN, two locally connected and three fully connected layers. They applied dropout with a rate of 0.5 and reached to the accuracy of 94% [32].

2.2. Sensor-Wise method

For any specific hand gesture, always some specific muscles are engaged. As a result of the hand gesture recognition, the contribution of these muscles should be determined. The contribution of

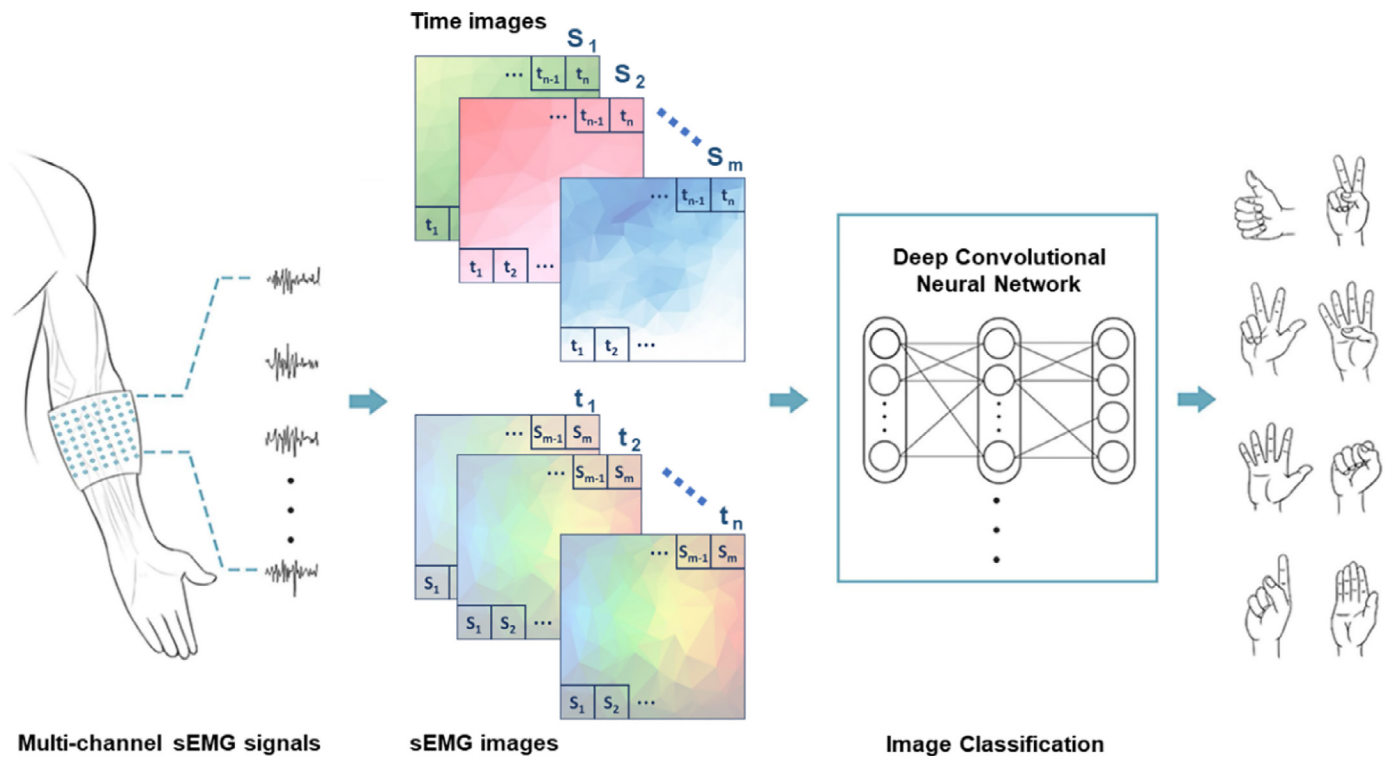


Fig. 1. a schematic illustration of gesture recognition for both types of images [21].

electrodes to an intended output can be defined through adjusting weights of the network, and we should look for patterns within the signal of each electrode over time. In the HD-sEMG image method, the patterns were extracted from the instantaneous values of electrodes. The HD-sEMG image by itself only contains spatial information, but the sequence of these images is interpreted to provide temporal information about changes over time. Because of the nature of sEMG signals and also the structure of CNN in extracting features, we introduced the time-image instead of the sEMG image in the proposed Sensor-Wise method. The time-images contain temporal information, but the sequence of these images is interpreted to provide spatial information. Both methods provide spatiotemporal information but in a different way. Fig. 1 shows the schematic illustration of gesture recognition for both types of images.

To explain the concept through a picture, we can consider that it has multiple color channels. Each color channel has a detectable pattern. But, if we put the n th pixel (pixel number n) of each channel next to the n th pixel of the other channels and create a new image, a meaningless image is obtained. Images in the sEMG image method are similar to these vague images. Putting these vague images together in the network makes the pattern extraction process more difficult, then the network will be heavier and likely suffer overfitting. The presence of multiple dropout layers indicates that the network has a significant overfitting problem [21, 26]. In this report, instead of training and then killing the neurons, we have used the pooling method to tune the number of parameters. The pooling method decreases the size of input data by applying mathematical or logical functions.

In the proposed Sensor-Wise method, each time-image corresponds to a particular sensor. We have feature channels, which are proportional to the number of electrodes. In this method, patterns are extracted in each time-image by the kernel, which is adjusted according to the nature of data. The relevance of patterns extracted from each electrode with other electrodes to perform a specific gesture is determined by adjusting the weights of the network. As

the images in this method are temporal information, the kernel is defined as one-dimensional. When using two-dimensional kernels, in addition to adjacent moments, other moments appear in the kernel that can make it difficult to extract patterns. Similar to [21, 33], we applied 256 samples (16×16 time-image), which is equivalent to 256-milliseconds.

Our neural network consists of two convolutional layers. The first hidden layer applies 32 filters with stride 1 to the input data, and the second hidden layer consists of 64 filters with stride 1. After each hidden layer, we applied ReLU activation and average pooling functions. The output layer indeed is a fully connected G-Way layer and softmax function where G is the number of gestures to be classified. We used Adam optimizer [34] with the learning rate of 5×10^{-4} and the decay factor of 10^{-6} .

To provide the input data to the network, a frame is considered for each electrode; each of these frames contains the values of the electrode at the present moment and the preceding ones. At each time of sampling from the electrodes, a new value enters the frame, and the oldest value is removed. This method is reasonable, because if we provide a large sample instead of smaller continuous samples for training the network, due to leading or lagging that may occur in the presence of a signal, the neural network may not be able to predict correctly over the signals which are produced from the same gesture.

3. Results

3.1. Evaluation with CapgMyo database

Our ConvNet algorithm is evaluated with the well-known CapgMyo DB-a database. HD-sEMG data was collected using a non-invasive wearable device consisting of 8 acquisition modules; each of them contains a matrix-type (2×8) differential electrode array. Each electrode has a diameter of 3 mm and was arranged with an interelectrode distance of 10.05 mm vertically and 7.5 mm horizontally. The EMG signals were sampled at 1 kHz, digitized with a 16-

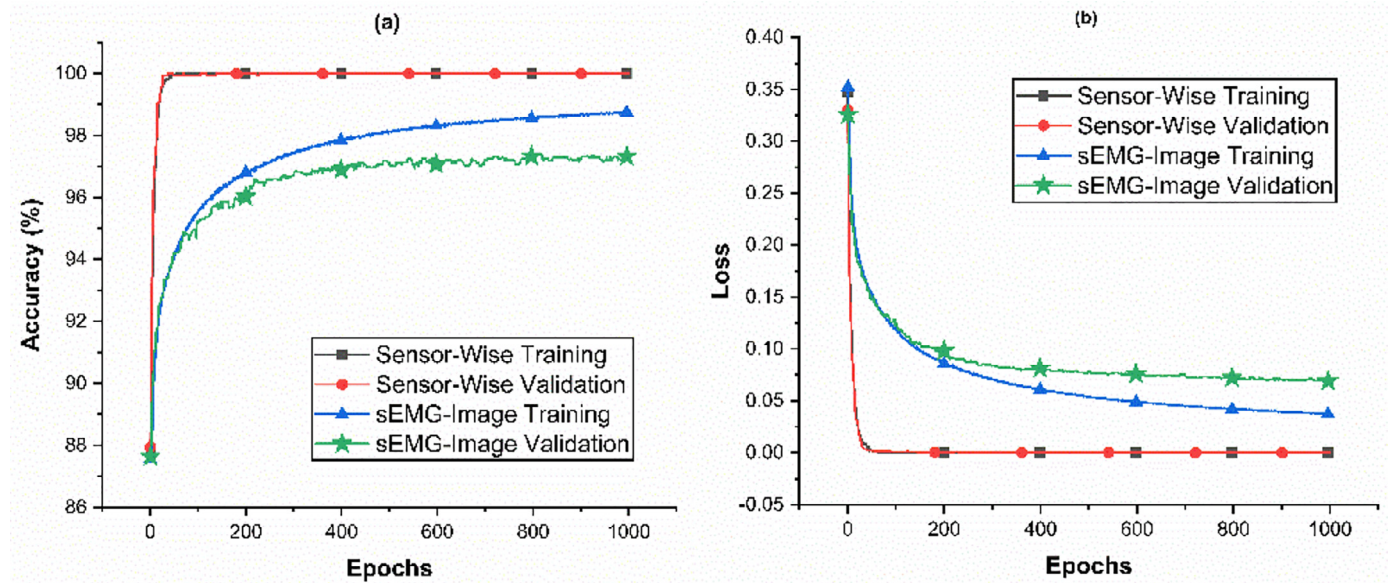


Fig. 2. Comparison of the sEMG-image method with the Sensor-Wise method. (a) Training and validation accuracy of both methods. (b) Training and validation loss of methods..

bit ADC and band-pass filtered at 20–380 Hz. The EMG data from all modules were packed in an ARM controller and transferred to a PC via WIFI. The database contains 8 isometric and isotonic dedicated hand gestures obtained from 18 subjects [21]. The data are rectified and divided by the maximum value in order to normalize it. The method is implemented and evaluated using the Keras and TensorFlow libraries on the Tesla K80 GPU.

In Section 3.1.1., we used two-dimensional layers to be able to compare the sEMG image and time-image. Due to the temporal nature of the images, the kernel is one-dimensional, then one-dimensional convolutional layers can be used instead of two-dimensional layers. Besides, applying one-dimensional images allows continuous scanning over time, while in two-dimensional images, the kernel stops at the end of each row. Therefore, in other evaluations, one-dimensional layers are applied. Kernel size for one-dimensional layers is 3, and the average pooling function with size 2 is applied after each of them. The accuracy of test-data is calculated as the proportion of known correct images and averaged over all the subjects. One of the trials is separated as test-set in turns, to conduct cross-validation, and training and validation are performed on the 9 remaining trials. One percent of the data is used as validation data during the training process. After evaluation, the network is trained over all 9 trials through 25 epochs, and then prediction over test data is made using simple majority voting [21].

3.1.1. Comparing sEMG image method with Sensor-Wise method

In this section, we compare the network performance based on time-image and SEMG image together. The Convolutional layers in this section are two-dimensional, with the kernel size of (1,3) and stride 1. After each convolutional layer, we applied (2,2) average pooling function. The input of the network for the Sensor-Wise method is 256 images with a size of 8×16 , and for the sEMG image, it is 128 images with a size of 16×16 . One percent of the input data is separated as validation data, and the network trained over the remaining data. Evaluation is done at the end of each training epoch. Fig. 2 shows the results of this test.

There is no sign of overfitting for both methods, while in previous studies, multiple layers of dropout used due to the significant overfitting problems [21, 26]. The Sensor-Wise method reached an

accuracy of 100% for both training and validation. But the sEMG image method even after 1000 epochs did not reach to the results in which obtained with the Sensor-Wise method. Besides, a greater gradient, presented in the training curve of the Sensor-Wise method indicates that patterns are recognized easier than the sEMG image method. The accuracy over test data reached to 99.98% for the Sensor-Wise method, and 97.32% for the sEMG image method.

3.1.2. Effect of the number of applied sensors on the accuracy of gesture recognition

Although the higher number of sensors in HD-sEMG provides better physiological information, gesture recognition may not require a large number of sensors. There should be an optimum number of sensors. The larger numbers of sensors require more hardware resources and more computational load. In the Sensor-Wise method, we can easily remove the feature (channel) of a sensor and investigate its effect on the performance of the network. There are eight columns, each containing 16 sensors. Table 1 shows the results of applying 128, 112 (col. 2 to 8), 96 (col. 3 to 8), 80 (col. 4 to 8), 64 (col. 5 to 8), 48 (col. 6 to 8), 32 (col. 7 and 8), and 16 (col. 8) electrodes on the network accuracy.

For each series of selected electrodes, the method reached the training accuracy of 100%. The validation accuracy of the method reached to 100% with up to 32 electrodes and 99.73% with 16 electrodes. For 128 to 64 electrodes, the test accuracy of 100% is obtained for 17 subjects (all subjects except subject 5). Using 48, 32, and 16 electrodes, the test accuracy of the network reached 100% over 13, 11, and 9 subjects, respectively. The accuracy obtained with just 16 electrodes is higher than the accuracy reported in previous studies [21, 26].

3.1.3. Effect of electrodes location along the forearm

The data obtained from various sensors in the HD-sEMG allows us to investigate the location effect of the sensors on system performance quantitatively. There are 8 columns of 16 electrodes in the CapgMyo DB-a database. Table 2 shows the results of applying each column separately. The highest accuracy is for columns 6 to 8, and the second and third columns have the least accuracy. This evaluation allows us to know which columns have the least effect on system performance.

Table 1

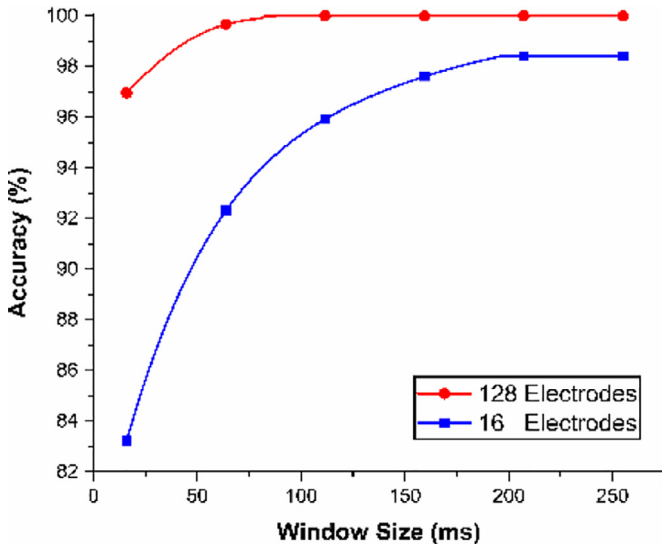
The effect of the number of applied sensors on the accuracy of gesture recognition.

	128	112	96	80	64	48	32	16
Training Accuracy	100%	100%	100%	100%	100%	100%	100%	100%
Training Loss	0	0	0	0	0	0	0	0
Validation Accuracy	100%	100%	100%	100%	100%	100%	100%	99.73%
Validation Loss	0	0	0	0	0	0	0	0.4
Test Accuracy	99.98%	99.98%	99.99%	99.97%	99.95%	99.94%	99.91%	99.78%

Table 2

The effect of electrodes' location along the forearm for each column of electrodes separately.

	Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7	Col. 8
Gesture 1	92.9%	88.7%	89.96%	94.22%	92.93%	95.38%	96.89%	99.66%
Gesture 2	96.7%	94.11%	98.67%	94.68%	94.73%	98.39%	99.34%	99.82%
Gesture 3	99.53%	94.37%	94.13%	99.83%	98.14%	99.96%	99.93%	99.53%
Gesture 4	90.54%	90.38%	88.05%	90.71%	90.63%	90.48%	93.54%	99.43%
Gesture 5	96.38%	91.64%	98.82%	95.16%	95.25%	99.42%	99.23%	99.97%
Gesture 6	99.38%	99.87%	99.54%	99.89%	93.59%	97.58%	99.02%	99.99%
Gesture 7	99.93%	98.3%	92.22%	97.62%	99.68%	99.06%	91.12%	99.97%
Gesture 8	98.16%	93.39%	91.51%	96.35%	98%	92.92%	96.05%	99.85%
Total	96.69%	93.84%	94.11%	96.05%	95.37%	96.65%	96.89%	99.78%

**Fig. 3.** The effect of window size in system performance for 128 and 16 electrodes.

3.1.4. Effect of window size in network performance

Instantaneous values of HD-sEMG are not reliable for accurate gesture recognition; then, we used time-images. After finalizing the number of electrodes as well as the network structure, we can look for the optimal size of the window. Fig. 3 indicates the results of applying different sizes of the window for 128 and 16 electrodes. In the evaluation with 16 electrodes with a window of 196 ms and with all 128 electrodes through a window of 100 ms, the network is able to achieve the same accuracy, which was obtained through a 256 ms window.

3.2. Evaluation with CSL-HDEMG database

We evaluated our method with the CSL-HDEMG database, which HD-sEMG signals of performing 27 finger gestures for 5 subjects recorded in 5 sessions over 10 trials. Bipolar sEMG signals were recorded through 192 electrodes at a sampling rate of 2048 Hz. We applied the same method of filtering and segmentation, as was mentioned in Amma et al. study [35]. Data are rectified and divided by the maximum value in order to normalize it. The method is implemented and evaluated using the Keras and

TensorFlow libraries on the Tesla K80 GPU. We separated one of the trials as the test-set in turns, to conduct cross-validation, and the classifier is trained using the 9 remaining trials. The network is trained through 25 epochs, and then prediction over test data is made using simple majority voting [21]. The result of the evaluation reached an accuracy of 99.55% through a window of 250 ms (512 samples).

4. Discussion

The results show that the Sensor-Wise method has a greater ability compared to the sEMG image method to extract the patterns. In the Sensor-Wise method, we find the patterns inside the signal of each electrode and contribution of different patterns in performing a specific gesture is defined by adjusting weights of the network, while in the sEMG image method, we look for patterns inside instantaneous values of different electrodes. Besides, due to the optimal structure of our method, there is no sign of overfitting, while in previous reports, multiple layers of dropout were applied in their networks due to significant overfitting problems [21, 26]. Instead of dropout layers, we used average pooling layers to adjust the number of parameters in the network.

Computational load is important when we want to implement an artificial intelligence network on real-time hardware like Field Programmable Gate Array (FPGA) [24]. Our proposed method just applies two convolutional hidden layers. Geng et al. applied a network with seven hidden layers, which consist of two convolutional, two locally connected, and three fully-connected layers along with eight batch normalizations and three dropout layers [21]. These extra layers, especially batch normalization functions, impose much computational load on the hardware. Hu et al. even added LSTM layers to the network introduced by Geng et al., that require even more hardware resources [26]. Our proposed model, due to its fewer computational load and the smaller size, is lightweight [36] compared to Geng et al. and Hu et al.

Furthermore, with 96 electrodes, the accuracy of the method reached almost 100% (99.99%). The better result with 96 sensors compared to 128 sensors could be due to removing the inappropriate signals caused by artifacts [37]. Geng et al. obtained an accuracy of 99.5%, and Hu et al. reached an accuracy of 99.7%, both using a window of 150 ms while, in this study, the accuracy of almost 100% is achieved using a window of just 100 ms. Also, it has the capability to reach the same accuracy even with the fewer number of electrodes. Even with 16 electrodes, the accuracy of

the method reached 99.8%, which is higher than the accuracies reported in the previous studies [21, 26].

Besides, the author believes that, by providing enough non-conflicting data, the accuracy of almost 100% could be achieved even with 32 electrodes as the accuracy reached 100% over more than half of subjects. We should consider that even in doing a simple gesture, different strength and speed, that a subject applies, can cause differences in the output signals [38]. We can only expect accurate predictions from the network when we provide enough data to train it.

Evaluation with the CSL-HDEMG database reached an accuracy of 99.55%, which is 2.75% higher than the accuracy reported by Geng et al. [21]. Amma et al., for the CSL-HDEMG database, due to the presence of artifacts and noise in the signals, in addition to the low pass filter, used the median filter, while Geng et al. for the CapgMyo DB-a database, just applied the low pass filter [21, 35]. Excessive noise and artifact in CSL-HDEMG database signals can be one of the reasons that the network did not reach the ultimate accuracy of about 100%.

5. Conclusions

In this study, we introduced the Sensor-Wise method, which has a high capability to extract features due to its high compatibility with the nature of sEMG signals and the structure of convolutional networks. The method is lightweight and has a lower computational load because it has an optimal size with only two hidden layers, and there are not any batch normalization functions in the network structure. The proposed method has shown no sign of overfitting, which is attributed to its lightweight structure and high compatibility and reached to the ultimate accuracy of almost 100%. Even with fewer sensors, it was able to reach a higher accuracy compared to reported results in the literature. The accurate and lightweight structure of this method make it a proper option for hardware implementation, especially in embedded real-time systems.

In future work, we plan to investigate sEMG data acquisition methods and develop a hardware-accelerated gesture recognition system. Besides, the proposed method provides an advanced approach for the conduction of studies on the gestures with more complexity and more fluid and natural human-computer interactions, especially for applications in rehabilitation or assistive robots.

Declaration of Competing Interest

None.

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